Practice Problems-5

An equiconvex lens located in air has radii of 5.20 cm, an index of 1.680, and a thickness of 3.50 cm. Calculate (*a*) the focal length and (*b*) the power of the lens. Find (*c*) the distances from the vertices to the focal points and (*d*) the principal points.

A glass lens with radii $r_1 = +2.50$ cm and $r_2 = +4.50$ cm has a thickness of 2.90 cm and an index of 1.630. Calculate (*a*) the focal length and (*b*) the power of the lens. Find the distances from the vertices to (*c*) the focal points and (*d*) the principal points.

A thick lens with radii $r_1 = -4.50$ cm and $r_2 = -3.60$ cm has a thickness of 3.0 cm and an index of 1.560. Calculate (*a*) the focal length and (*b*) the power of the lens. Find also the distances from the vertices to the corresponding (*c*) focal points and (*d*) principal points.

A glass lens 4.50 cm thick, and index 1.70, has radii of $r_1 = +3.0$ cm and $r_2 = +3.50$ cm. If a liquid of index 1.320 is in contact with r_1 and a very dense, transparent, oil of index 2.20 is in contact with r_2, find (*a*) the primary and secondary focal lengths and (*b*) the power of this optical system. Also find the distances from the two vertices to (*c*) the principal points, (*d*) the focal points, and (*e*) the nodal points. (*f*) If an object is located in the liquid of index 1.320 and 13.50 cm from r_1, find the image position.

Solve Prob. 5.17 graphically, locating the six cardinal points of the lens system and the image distance.